5 A doctor is investigating the concentration of blood glucose in patients at risk of developing type 2 diabetes, where blood glucose is measured in appropriate units. The doctor claims that a particular intervention reduces the concentration by more than k units on average. A group of 8 at risk patients is selected at random and each patient follows the intervention for six months. The blood glucose concentrations before and after the intervention are given in the following table.

Patient	*A*	*B*	*C*	*D*	*E*	*F*	*G*	*H*
Before	183	165	172	165	143	176	161	153
After	164	148	164	149	134	153	155	148

(a) Use a t-test at the 5% significance level to find the range of values of k for which the result of the test is to reject the null hypothesis. [7]

(b) State an assumption necessary for the test in part **(a)** to be valid. [1]